

Batch: A1 Roll No.: 16010123012**Experiment No.: 6****Grade: AA / AB / BB / BC / CC / CD / DD****Title:** Implementing Database as a Service on Azure Cloud and performing CRUD operations on the database.**Objective:**

To create and configure a Database as a Service (DBaaS) on the Microsoft Azure cloud platform and perform CRUD (Create, Read, Update, Delete) operations on the database using a client application or query interface.

Expected Outcome of Experiment:

CO	Outcome
CO4	Build cloud services and applications

Books/ Journals/ Websites referred:

<https://learn.microsoft.com/en-us/azure/azure-sql/database/single-database-create-quickstart?view=azuresql&tabs=azure-portal>
<https://learn.microsoft.com/en-us/azure/azure-sql/database/>

Related Theory: -

Database as a Service (DBaaS) is a cloud computing service model in which the cloud provider offers fully managed database services over the internet. Users can create, configure, and use databases without worrying about hardware provisioning, operating systems, or database server maintenance. In this service the cloud provider manages infrastructure, OS, database engine installation, patching, backups, and availability. The consumer focuses only on data, schema design, and queries.

Key Components of DBaaS

- 1. Database Engine**
 - Azure SQL Database (relational DBMS)
- 2. Storage Management**
 - Automatic data storage and scaling
 - Built-in backup and recovery
- 3. Networking**
 - Secure connectivity via firewall rules

- Public/private access configuration
- 4. **Management Interface**
 - Azure Portal
 - SQL Query Editor
 - Client tools (SSMS)
- 5. **Security & Availability**
 - Authentication & authorization
 - High availability and fault tolerance

Responsibility Division in DBaaS:

Layer	Managed By
Physical Data Center	Cloud Provider
Servers & Storage	Cloud Provider
Operating System	Cloud Provider
Database Engine	Cloud Provider
Backup & Patching	Cloud Provider
Database Schema & Data	Consumer
Queries & Applications	Consumer

Characteristics of DBaaS

1. On-Demand Provisioning
Databases can be created instantly through the Azure portal without manual setup.
2. Scalability
Database storage and performance can be scaled up or down based on workload.
3. Managed Maintenance
Patching, updates, backups, and monitoring are handled automatically.
4. High Availability
DBaaS ensures minimal downtime using replication and fault-tolerant architecture.

5. Pay-as-You-Use

Users are billed only for the database resources they consume.

Implementation Details

Follow the steps provided in the reference link or Azure documentation:

1. Create an **Azure SQL Database** using the Azure Portal.
2. Configure database server settings (authentication, firewall rules).
3. Connect to the database using:
 - Azure Query Editor or
 - SQL Server Management Studio (SSMS).
4. Create a sample table in the database.
5. Perform **CRUD operations**:
 - **Create** – Insert records
 - **Read** – Retrieve records
 - **Update** – Modify existing records
 - **Delete** – Remove records

Sample CRUD Operations:

1. -- Create

```
INSERT INTO Student VALUES (1, 'Amit', 21);
```

2. -- Read

```
SELECT * FROM Student;
```

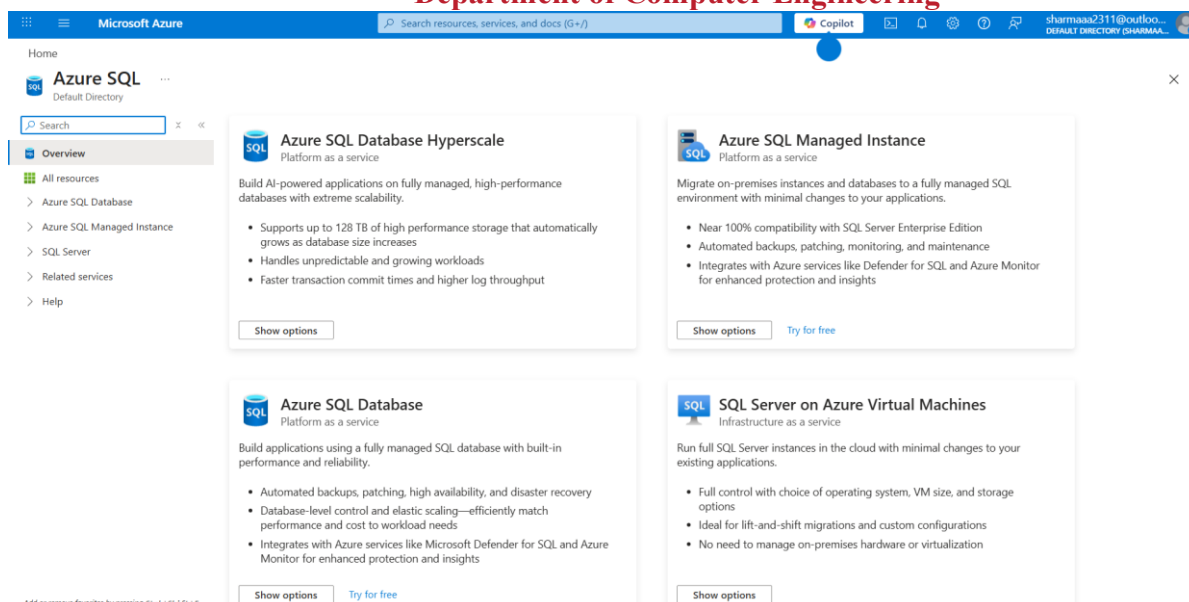
3. Update

```
UPDATE Student SET Age = 22 WHERE ID = 1;
```

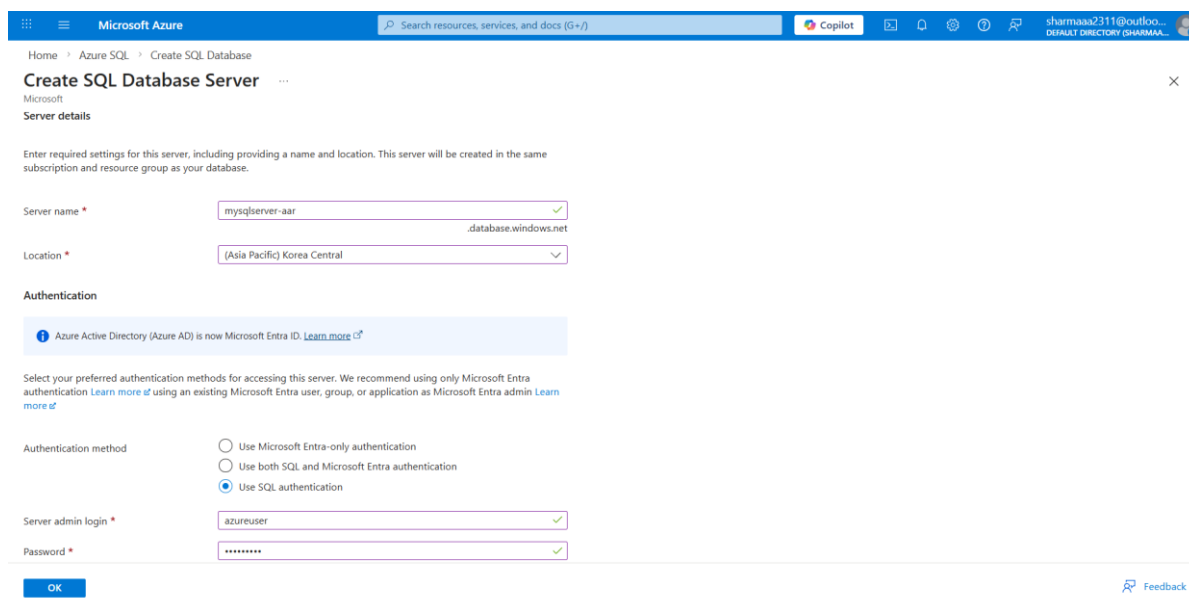
4. Delete

```
DELETE FROM Student WHERE ID = 1;
```

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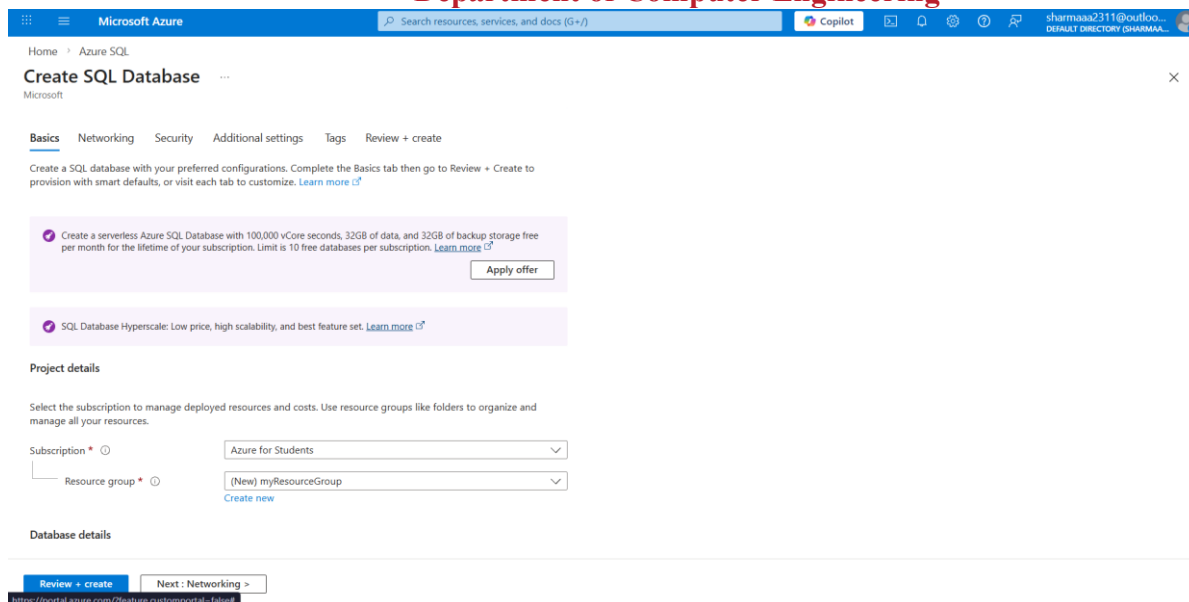


Azure Portal home page showing navigation panel for creating cloud resources

The screenshot shows the 'Create SQL Database Server' configuration page. It includes a breadcrumb trail: 'Home > Azure SQL > Create SQL Database'. The page is titled 'Create SQL Database Server' and has a 'Server details' section. It contains several form fields: 'Server name' (with a dropdown menu showing 'mysqlserver-aar'), 'Location' (with a dropdown menu showing '(Asia Pacific) Korea Central'), 'Authentication' (with a note about Azure Active Directory), 'Authentication method' (with radio buttons for 'Use Microsoft Entra-only authentication', 'Use both SQL and Microsoft Entra authentication', and 'Use SQL authentication'), 'Server admin login' (with a text field showing 'azureuser'), and 'Password' (with a masked text field). There is an 'OK' button at the bottom left and a 'Feedback' link at the bottom right.

Configuration of a new Database

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Home > Azure SQL

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Create a serverless Azure SQL Database with 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of your subscription. Limit is 10 free databases per subscription. [Learn more](#)

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * (New) myResourceGroup

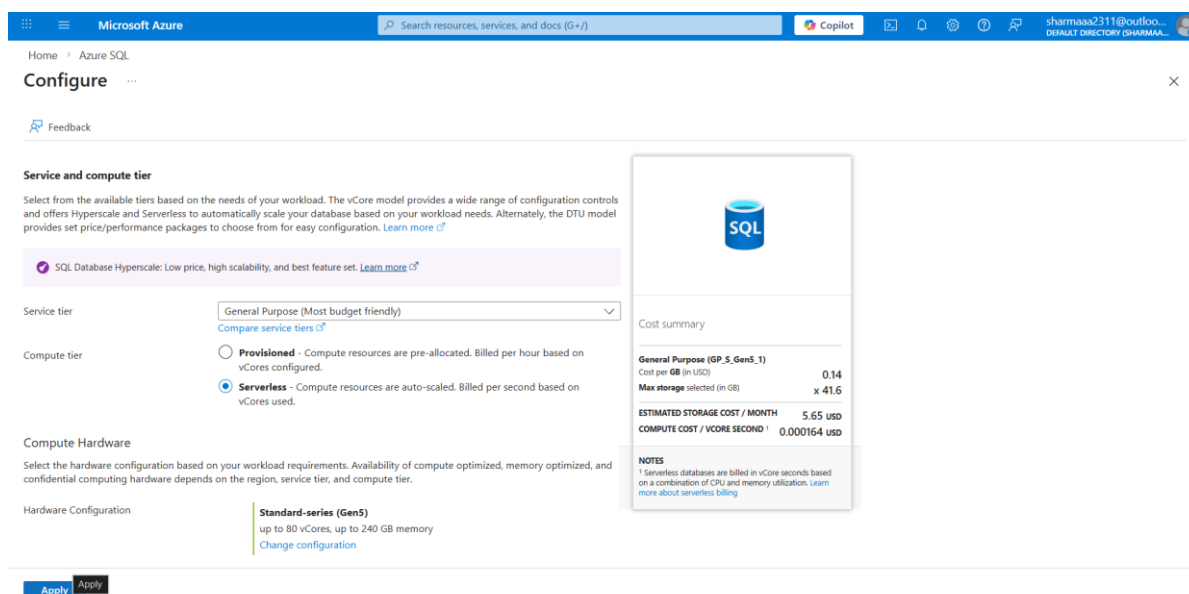
[Create new](#)

Database details

[Review + create](#) [Next: Networking >](#)

<https://portal.azure.com/#feature/customportal-false&>

Configuration of a new Database



Home > Azure SQL

Configure

Feedback

Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier General Purpose (Most budget friendly)

Compare service tiers

Compute tier

☐ Provisioned - Compute resources are pre-allocated. Billed per hour based on vCores configured.

☒ Serverless - Compute resources are auto-scaled. Billed per second based on vCores used.

Compute Hardware

Select the hardware configuration based on your workload requirements. Availability of compute optimized, memory optimized, and confidential computing hardware depends on the region, service tier, and compute tier.

Hardware Configuration

Standard-series (Gen5)

up to 80 vCores, up to 240 GB memory

[Change configuration](#)

Cost summary

General Purpose (GP_S_Gen5_1)

Cost per GB (in USD) 0.14

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH 5.65 USD

COMPUTE COST / VCORE SECOND ¹ 0.000164 USD

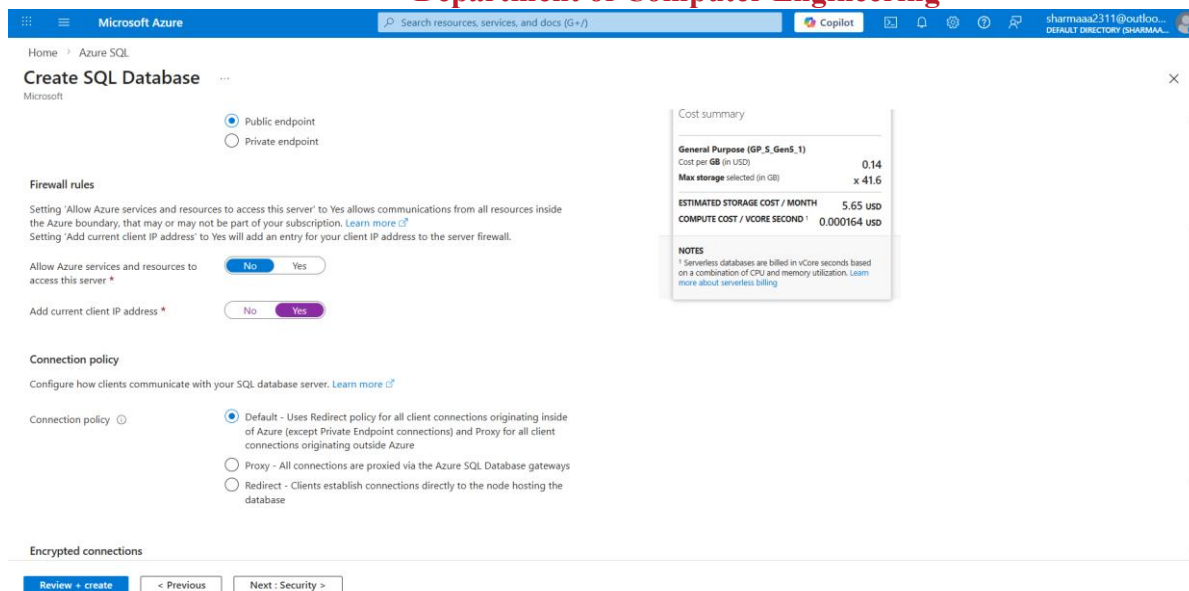
NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

[Apply](#) [Apply](#)

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Home > Azure SQL

Create SQL Database

Microsoft

☒ Public endpoint
☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

☐ No ☒ Yes

Add current client IP address *

☐ No ☒ Yes

Connection policy

Configure how clients communicate with your SQL database server. [Learn more](#)

Connection policy

☒ Default - Uses Redirect policy for all client connections originating inside of Azure (except Private Endpoint connections) and Proxy for all client connections originating outside Azure

☐ Proxy - All connections are proxied via the Azure SQL Database gateways

☐ Redirect - Clients establish connections directly to the node hosting the database

Encrypted connections

[Review + create](#) [< Previous](#) [Next : Security >](#)

Cost summary

General Purpose (GP_S_Gen5_1)

Cost per GB (in USD) 0.14

Max storage selected (in GB) x 41.6

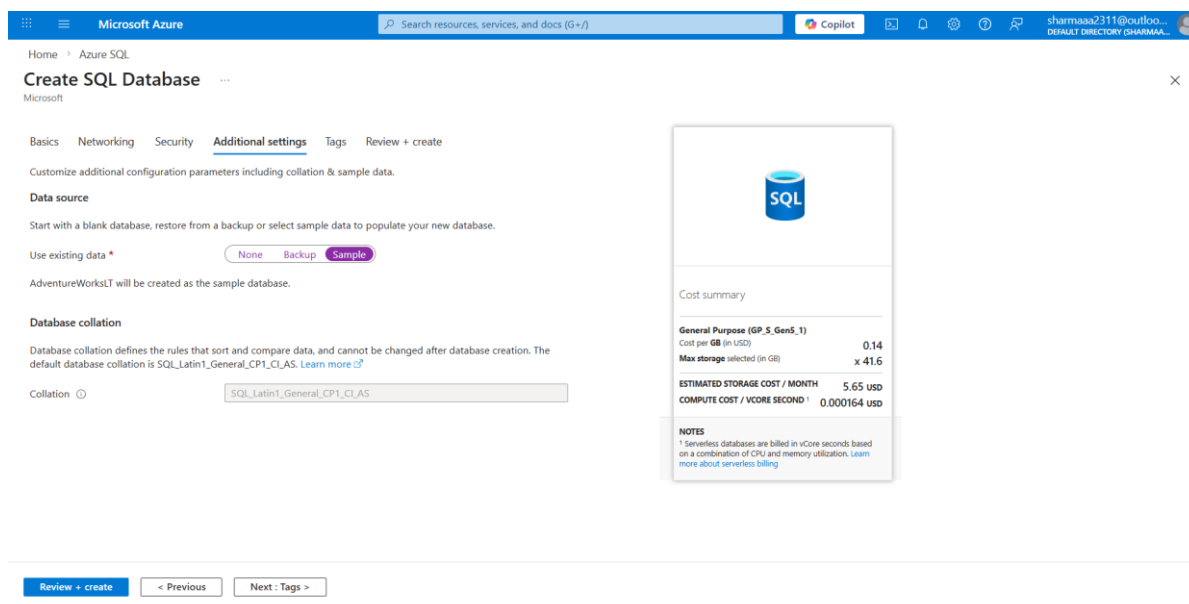
ESTIMATED STORAGE COST / MONTH 5.65 USD

COMPUTE COST / VCORE SECOND ¹ 0.000164 USD

NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

Configuration of a new Database



Home > Azure SQL

Create SQL Database

Microsoft

Basics Networking Security **Additional settings** Tags Review + create

Customize additional configuration parameters including collation & sample data.

Data source

Start with a blank database, restore from a backup or select sample data to populate your new database.

Use existing data *

☐ None ☐ Backup ☒ Sample

AdventureWorksLT will be created as the sample database.

Database collation

Database collation defines the rules that sort and compare data, and cannot be changed after database creation. The default database collation is SQL_Latin1_General_CP1_CI_AS. [Learn more](#)

Collation

SQL_Latin1_General_CP1_CI_AS

Cost summary

General Purpose (GP_S_Gen5_1)

Cost per GB (in USD) 0.14

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH 5.65 USD

COMPUTE COST / VCORE SECOND ¹ 0.000164 USD

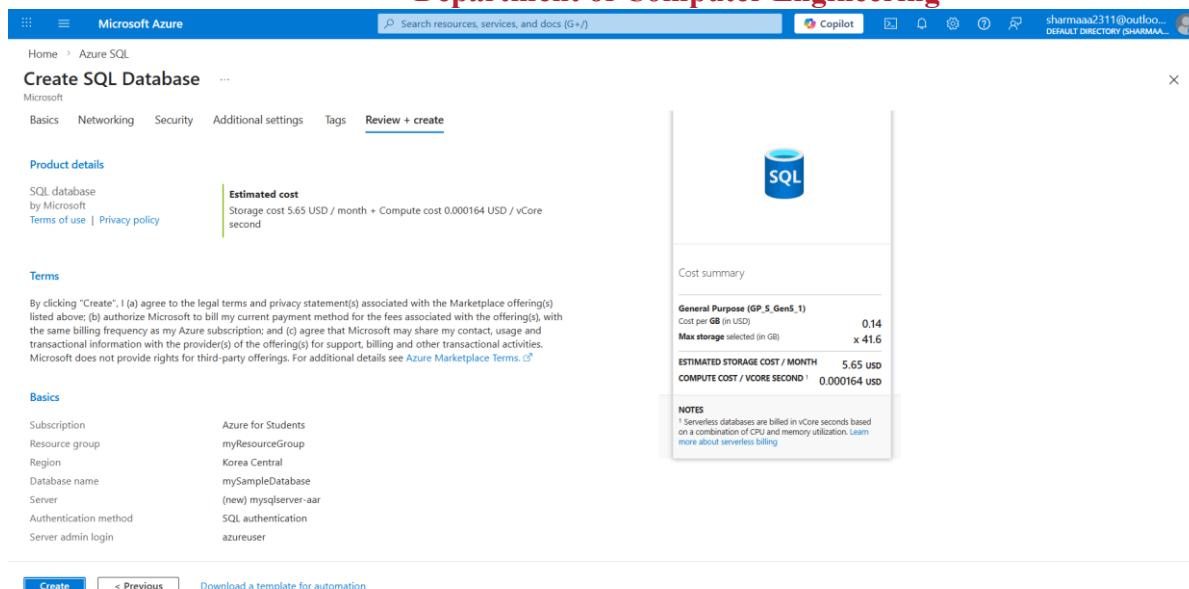
NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

[Review + create](#) [< Previous](#) [Next : Tags >](#)

Configuration of a new Database

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The screenshot shows the 'Create SQL Database' wizard in the Microsoft Azure portal. The 'Review + create' tab is selected. The 'Estimated cost' section shows a storage cost of 5.65 USD / month and a compute cost of 0.000164 USD / vCore second. The 'Cost summary' table is as follows:

Cost summary	
General Purpose (GP_S_Gen5_1)	
Cost per GB (in USD)	0.14
Max storage selected (in GB)	x 41.6
ESTIMATED STORAGE COST / MONTH	5.65 USD
COMPUTE COST / VCORE SECOND¹	0.000164 USD

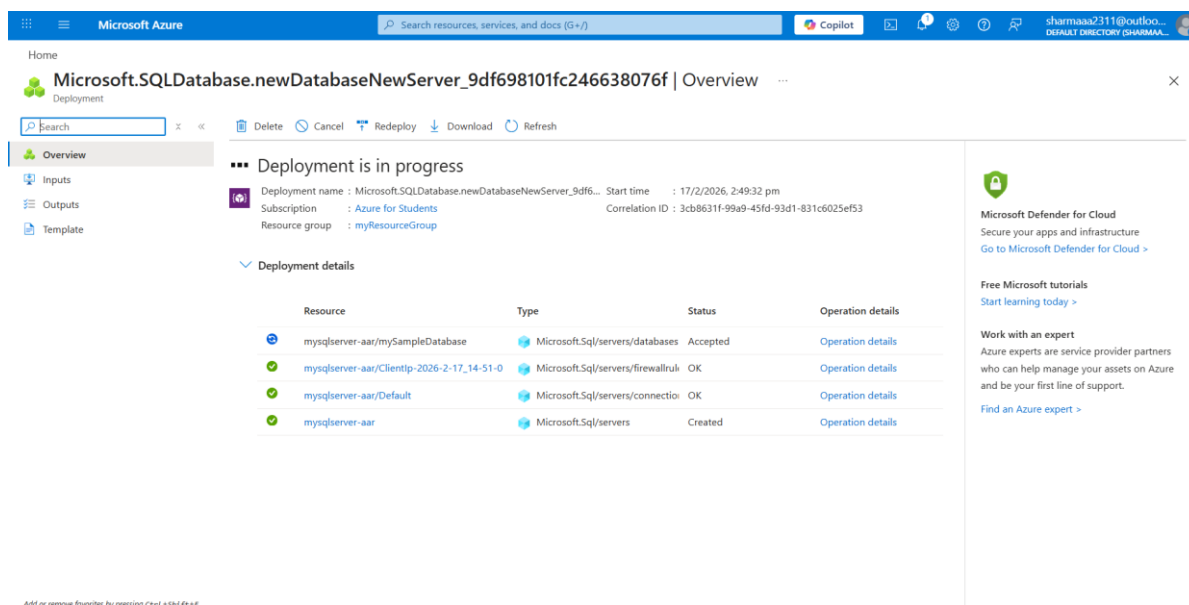
NOTES
¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

The 'Basics' section shows the following configuration:

- Subscription: Azure for Students
- Resource group: myResourceGroup
- Region: Korea Central
- Database name: mySampleDatabase
- Server: (new) mysqlserver-aar
- Authentication method: SQL authentication
- Server admin login: azureuser

Buttons at the bottom: **Create**, **< Previous**, and [Download a template for automation](#).

Validation of database configuration before deployment



The screenshot shows the 'Deployment is in progress' overview for the resource Microsoft.SQLDatabase.newDatabaseNewServer_9df698101fc246638076f. The deployment name is Microsoft.SQLDatabase.newDatabaseNewServer_9df6... and the start time is 17/2/2026, 2:49:32 pm. The correlation ID is 3cb8631f-99a9-45fd-93d1-831c6025ef53.

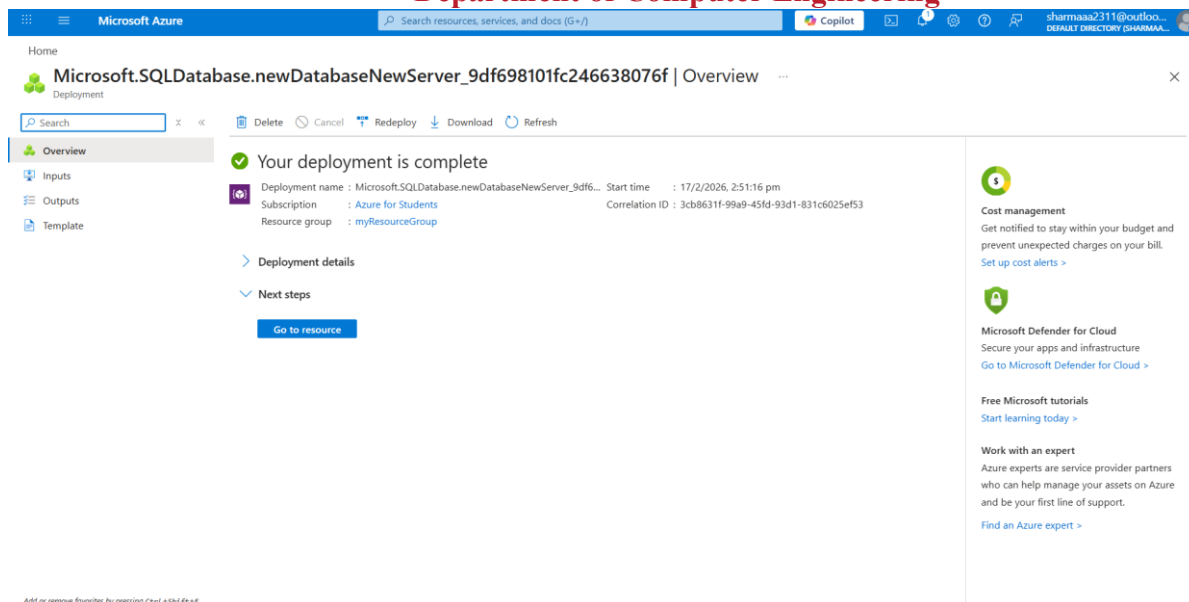
Deployment details

Resource	Type	Status	Operation details
mysqlserver-aar/mySampleDatabase	Microsoft.Sql/servers/databases	Accepted	Operation details
mysqlserver-aar/ClientIp-2026-2-17_14-51-0	Microsoft.Sql/servers/firewallrul	OK	Operation details
mysqlserver-aar/Default	Microsoft.Sql/servers/connection	OK	Operation details
mysqlserver-aar	Microsoft.Sql/servers	Created	Operation details

On the right, there are links for **Microsoft Defender for Cloud**, **Free Microsoft tutorials**, and **Work with an expert**.

Creation of the database

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Home

Microsoft Azure

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Copilot

sharmaa2311@outloo...
DEFAULT DIRECTORY (SHARMAA...)

Microsoft.SQLDatabase.newDatabaseNewServer_9df698101fc246638076f | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_9df6... Start time : 17/2/2026, 2:51:16 pm
Subscription : Azure for Students Correlation ID : 3cb8631f-99a9-45fd-93d1-831c6025ef53
Resource group : myResourceGroup

Deployment details

Next steps

[Go to resource](#)

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

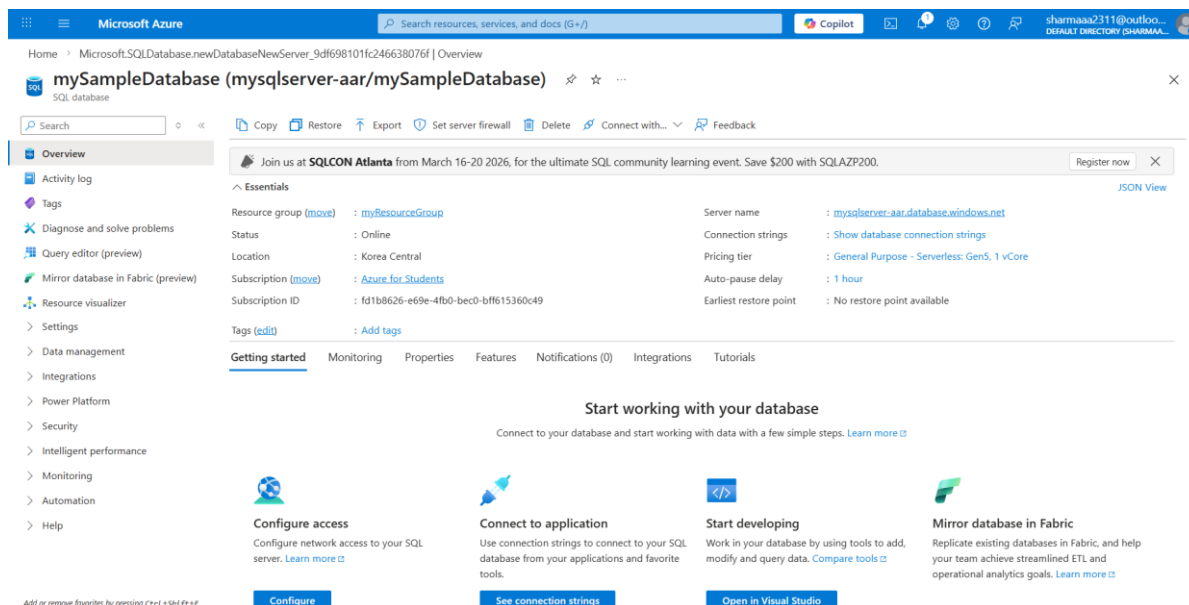
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Overview page of the deployed Database



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DEFAULT DIRECTORY (SHARMAA...)

Microsoft.SQLDatabase.newDatabaseNewServer_9df698101fc246638076f | Overview

mySampleDatabase (mysqlserver-aar/mySampleDatabase)

SQL database

Search

Copy Restore Export Set server firewall Delete Connect with... Feedback

Join us at **SQLCON Atlanta** from March 16-20 2026, for the ultimate SQL community learning event. Save \$200 with SQLAZP200. [Register now](#)

Essentials

Resource group (move) : myResourceGroup

Status : Online

Location : Korea Central

Subscription (move) : Azure for Students

Subscription ID : fd1b8626-e69e-4fb0-bec0-bff615360c49

Tags (edit) : Add tags

Server name : mysqlserver-aar.database.windows.net

Connection strings : [Show database connection strings](#)

Pricing tier : General Purpose - Serverless: Gen5, 1 vCore

Auto-pause delay : 1 hour

Earliest restore point : No restore point available

Getting started Monitoring Properties Features Notifications (0) Integrations Tutorials

Start working with your database
Connect to your database and start working with data with a few simple steps. [Learn more](#)

Configure access
Configure network access to your SQL server. [Learn more](#)

Connect to application
Use connection strings to connect to your SQL database from your applications and favorite tools. [See connection strings](#)

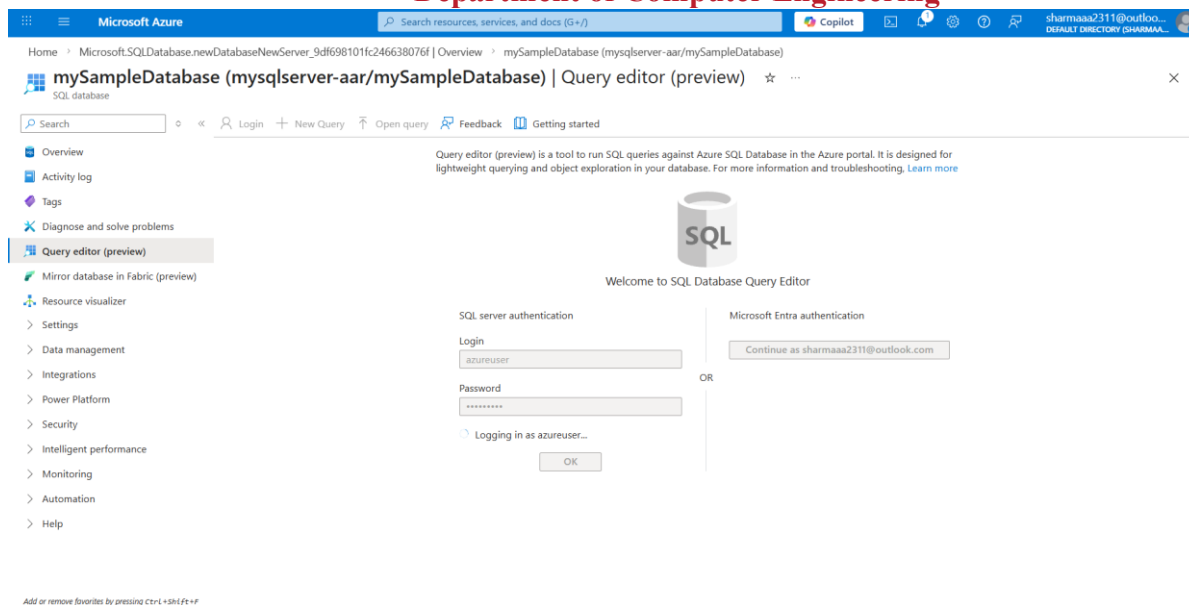
Start developing
Work in your database by using tools to add, modify and query data. [Compare tools](#)

Mirror database in Fabric
Replicate existing databases in Fabric, and help your team achieve streamlined ETL and operational analytics goals. [Learn more](#)

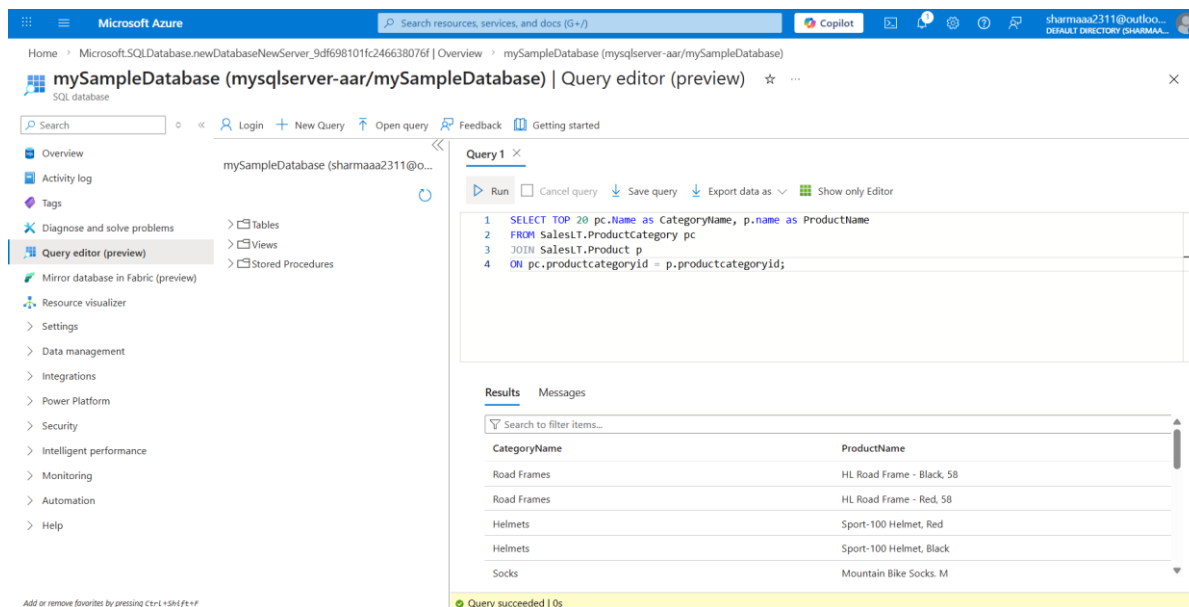
Add or remove favorites by pressing Ctrl+Shift+F

My Database

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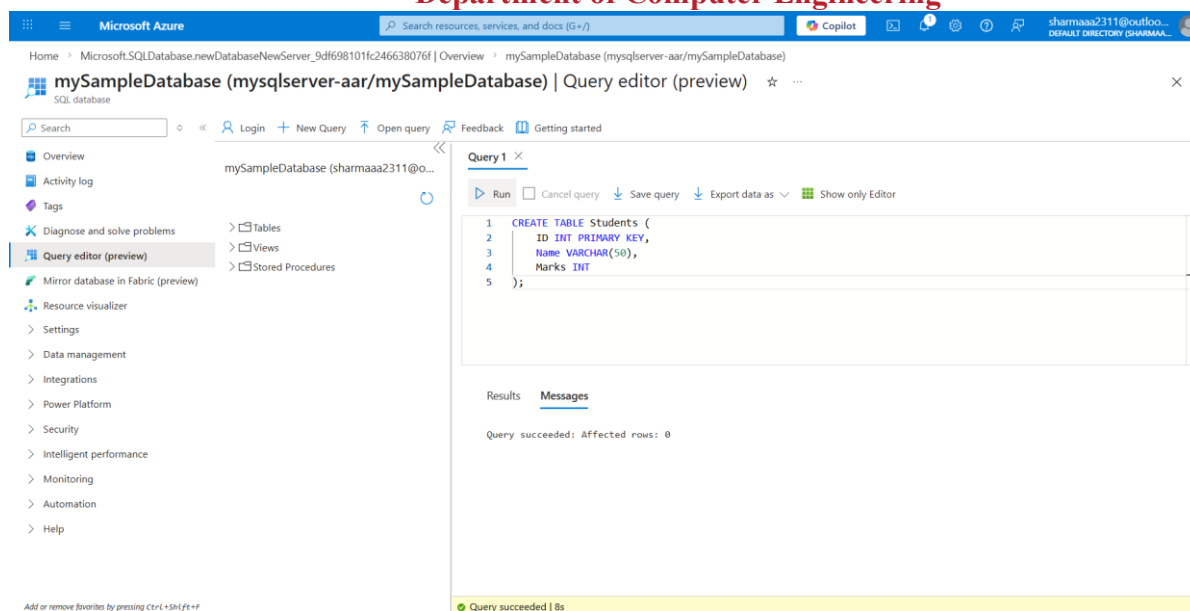


Authentication screen for Database Query Editor



SQL Query Editor connected to the Database for executing commands

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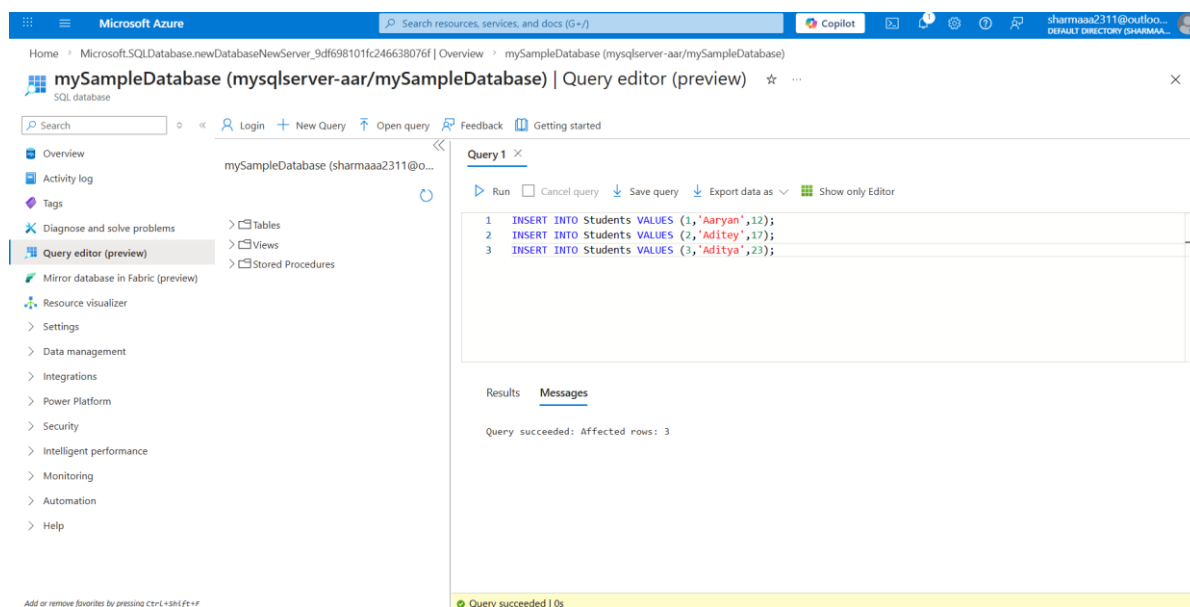
The screenshot shows the Microsoft Azure portal interface for a SQL database named 'mySampleDatabase'. The left sidebar contains navigation options like Overview, Activity log, Tags, and Query editor (preview). The main area displays the 'Query editor (preview)' for 'mySampleDatabase (sharmaa2311@o...)'.

The query editor shows the following SQL statement:

```
1 CREATE TABLE Students (  
2   ID INT PRIMARY KEY,  
3   Name VARCHAR(50),  
4   Marks INT  
5 );
```

The 'Results' tab is selected, showing 'Query succeeded: Affected rows: 0'. A status bar at the bottom indicates 'Query succeeded | 8s'.

SQL statement execution



The screenshot shows the Microsoft Azure portal interface for a SQL database named 'mySampleDatabase'. The left sidebar contains navigation options like Overview, Activity log, Tags, and Query editor (preview). The main area displays the 'Query editor (preview)' for 'mySampleDatabase (sharmaa2311@o...)'.

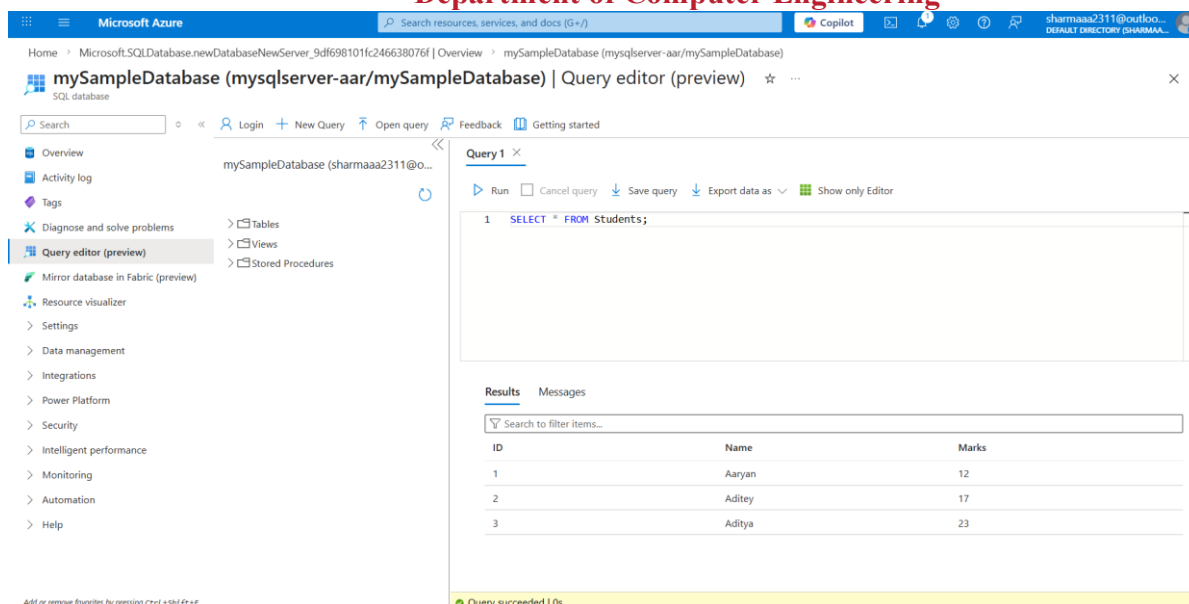
The query editor shows the following SQL statement:

```
1 INSERT INTO Students VALUES (1,'Aaryan',12);  
2 INSERT INTO Students VALUES (2,'Aditey',17);  
3 INSERT INTO Students VALUES (3,'Aditya',23);
```

The 'Results' tab is selected, showing 'Query succeeded: Affected rows: 3'. A status bar at the bottom indicates 'Query succeeded | 0s'.

SQL statement execution

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Microsoft Azure | Search resources, services, and docs (G+I) | Copilot | sharmaa2311@outloo... | DEFAULT DIRECTORY (SHARMAA...)

Home > Microsoft.SQLDatabase.newDatabaseNewServer_9df698101fc246638076f | Overview > mySampleDatabase (mysqlserver-aar/mySampleDatabase)

mySampleDatabase (mysqlserver-aar/mySampleDatabase) | Query editor (preview) ☆ ...

SQL database

Search < Login + New Query Open query Feedback Getting started

Overview
Activity log
Tags
Diagnose and solve problems > Tables
Query editor (preview) > Views
Mirror database in Fabric (preview) > Stored Procedures
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation
Help

mySampleDatabase (sharmaa2311@o...

Query 1 X

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT * FROM Students;
```

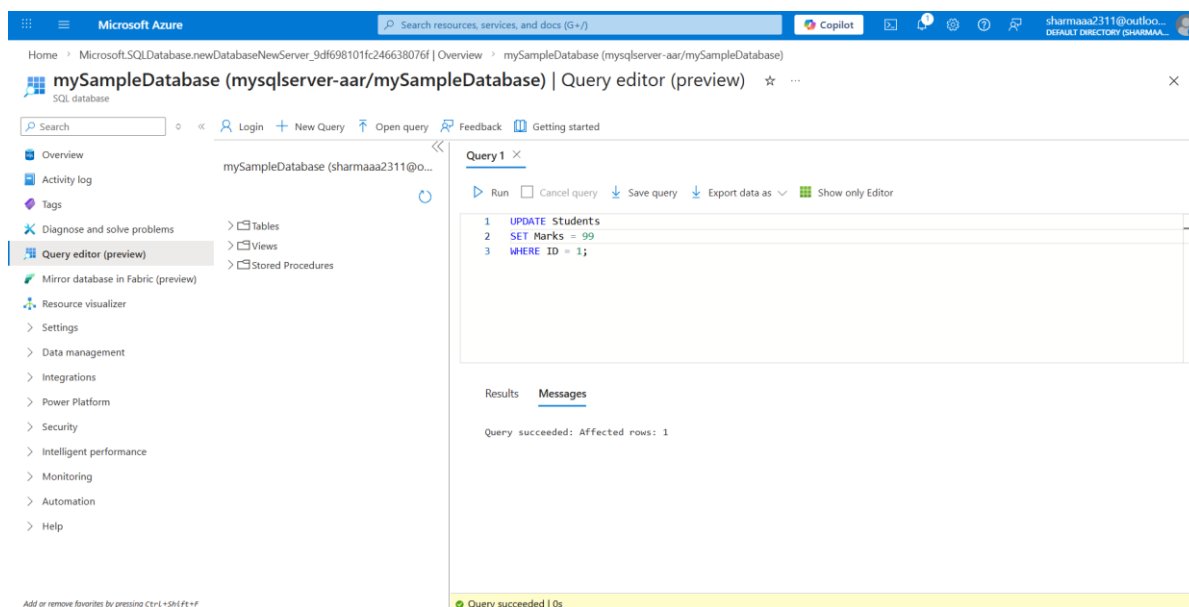
Results Messages

Search to filter items...

ID	Name	Marks
1	Aaryan	12
2	Aditey	17
3	Aditya	23

Query succeeded | 0s

SQL statement execution



Microsoft Azure | Search resources, services, and docs (G+I) | Copilot | sharmaa2311@outloo... | DEFAULT DIRECTORY (SHARMAA...)

Home > Microsoft.SQLDatabase.newDatabaseNewServer_9df698101fc246638076f | Overview > mySampleDatabase (mysqlserver-aar/mySampleDatabase)

mySampleDatabase (mysqlserver-aar/mySampleDatabase) | Query editor (preview) ☆ ...

SQL database

Search < Login + New Query Open query Feedback Getting started

Overview
Activity log
Tags
Diagnose and solve problems > Tables
Query editor (preview) > Views
Mirror database in Fabric (preview) > Stored Procedures
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation
Help

mySampleDatabase (sharmaa2311@o...

Query 1 X

Run Cancel query Save query Export data as Show only Editor

```
1 UPDATE Students
2 SET Marks = 99
3 WHERE ID = 1;
```

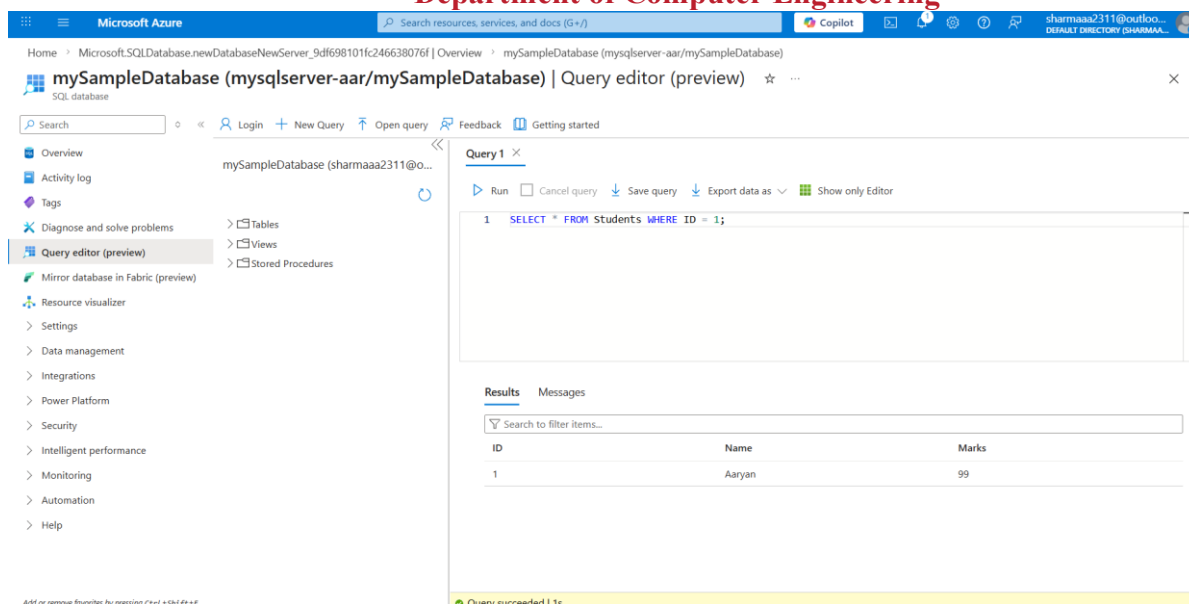
Results Messages

Query succeeded: Affected rows: 1

Query succeeded | 0s

SQL statement execution

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Microsoft Azure portal interface showing the execution of a SQL query in the Query editor (preview) for the mySampleDatabase (mysqlserver-aar/mySampleDatabase).

The query executed is:

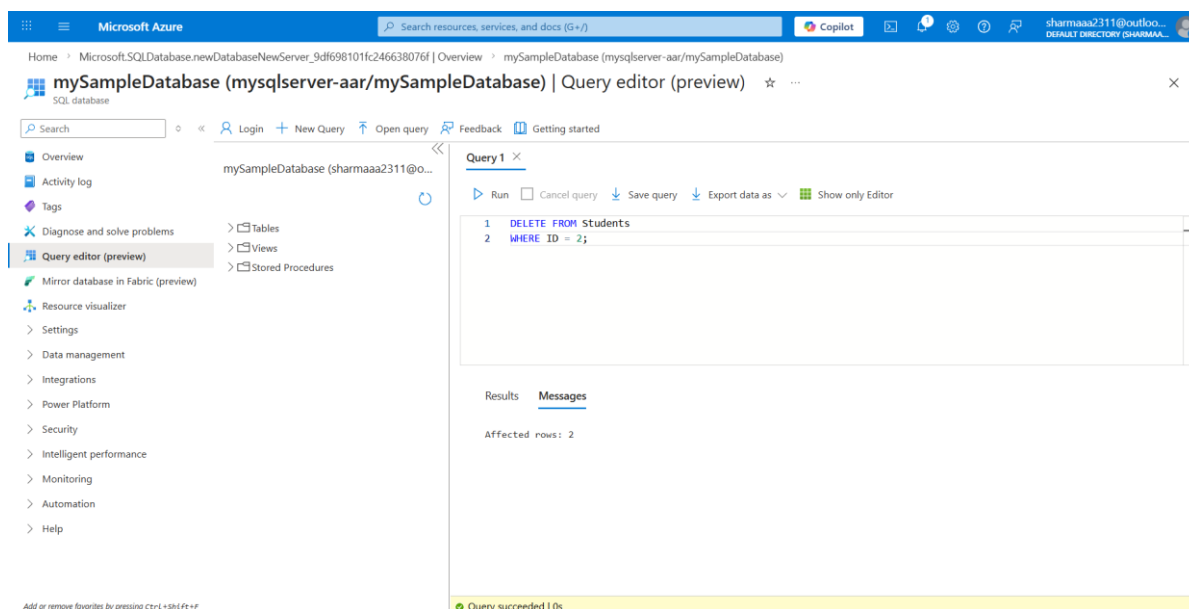
```
SELECT * FROM Students WHERE ID = 1;
```

The results are displayed in a table:

ID	Name	Marks
1	Aaryan	99

Query succeeded | 1s

SQL statement execution



Microsoft Azure portal interface showing the execution of a SQL query in the Query editor (preview) for the mySampleDatabase (mysqlserver-aar/mySampleDatabase).

The query executed is:

```
DELETE FROM Students  
WHERE ID = 2;
```

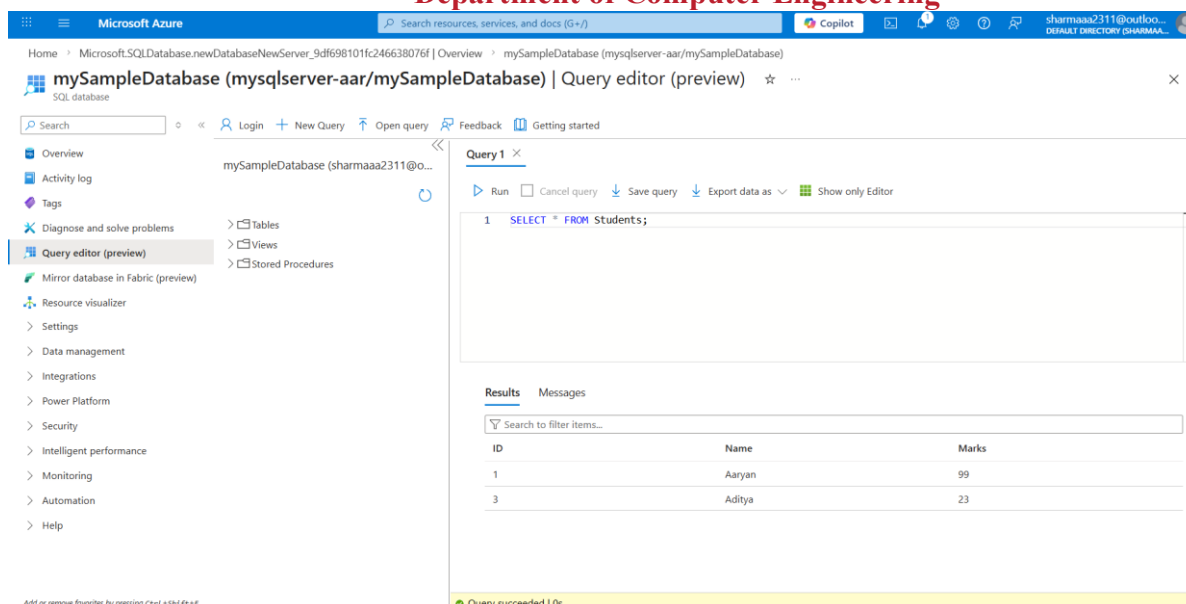
The results are displayed in a table:

Affected rows: 2		

Query succeeded | 0s

SQL statement execution

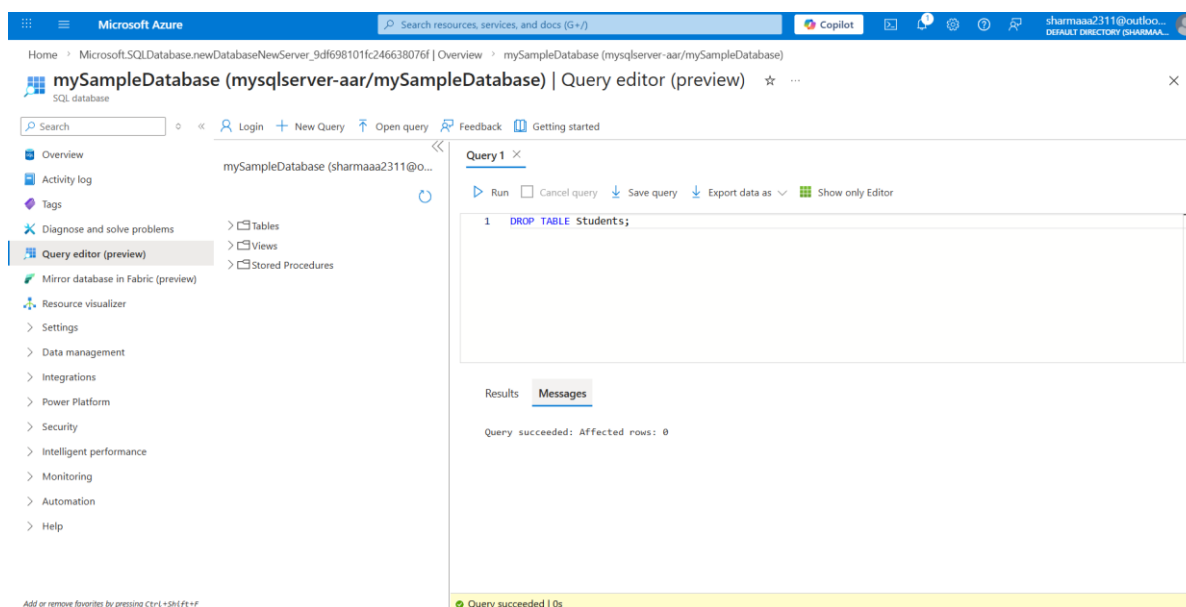
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The screenshot shows the Microsoft Azure portal interface for a SQL database named 'mySampleDatabase'. The 'Query editor (preview)' tab is active, displaying a query: `SELECT * FROM Students;`. The 'Results' tab is selected, showing a table with 2 rows and 3 columns: ID, Name, and Marks. The first row has ID 1, Name Aaryan, and Marks 99. The second row has ID 3, Name Aditya, and Marks 23. A status bar at the bottom indicates 'Query succeeded | 0s'.

ID	Name	Marks
1	Aaryan	99
3	Aditya	23

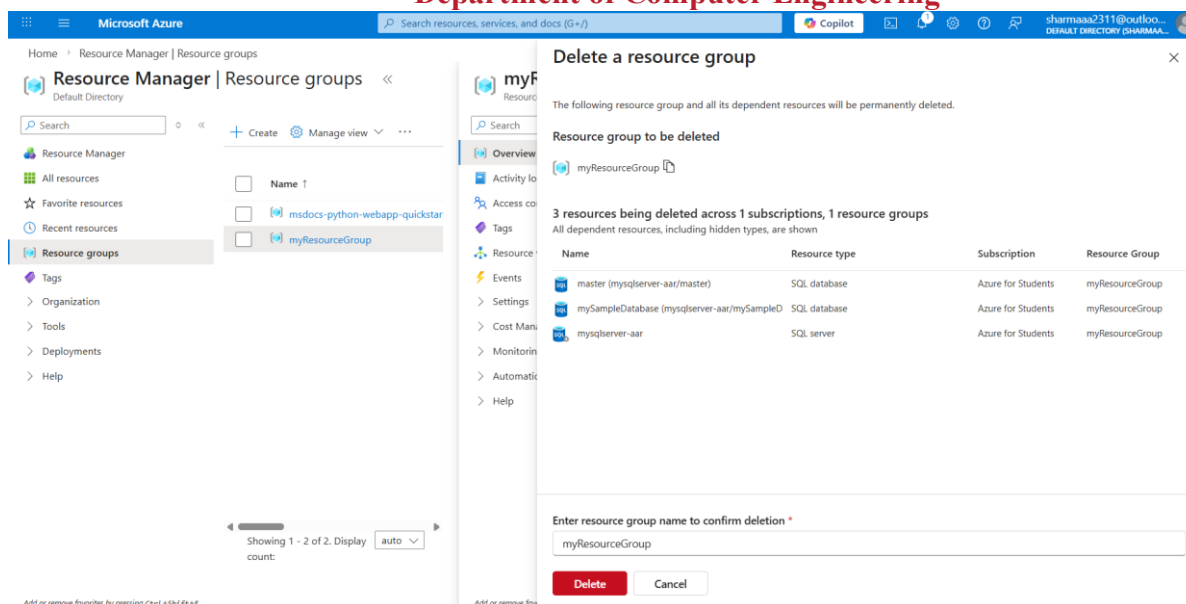
SQL statement execution



The screenshot shows the Microsoft Azure portal interface for the same SQL database. The 'Query editor (preview)' tab is active, displaying a query: `DROP TABLE Students;`. The 'Messages' tab is selected, showing the message: 'Query succeeded: Affected rows: 0'. A status bar at the bottom indicates 'Query succeeded | 0s'.

SQL statement execution

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Deleting the service

Post Lab Questions:

Q1. How does Database as a Service (DBaaS) reduce administrative overhead compared to managing an on-premise database server?

Database as a Service significantly reduces administrative overhead because most infrastructure and maintenance tasks are handled by the cloud provider. In an on-premise database setup, administrators must procure hardware, install the operating system and database software, configure networking, apply security patches, monitor performance, perform backups, and ensure fault tolerance. In DBaaS, tasks such as provisioning, scaling, patch management, backup scheduling, recovery, monitoring, and high availability are automated and managed by Azure. This allows developers and organizations to focus on database design, data management, and application logic rather than system administration, thereby improving productivity and reducing operational complexity.

Q2. Why is DBaaS preferred for applications requiring high availability and automatic backup?

DBaaS platforms like Azure SQL Database are designed with built-in mechanisms for high availability and data protection. The service uses data replication, fault-tolerant storage, and automated failover to ensure minimal downtime even during hardware or software failures. Additionally, automatic backups are performed regularly without manual intervention, enabling point-in-time recovery in case of accidental deletion, corruption, or system failure. Because these capabilities are managed by the cloud provider, applications can achieve enterprise-level reliability and disaster recovery without deploying complex backup systems or redundant infrastructure. Therefore, DBaaS is preferred for mission-critical applications where continuous availability and data safety are essential.

Conclusion:

In this experiment, a Database as a Service (DBaaS) instance was successfully created on the Microsoft Azure cloud platform using Azure SQL Database. The database server was configured with appropriate authentication and firewall settings, and connectivity was established using a query interface. CRUD operations were performed on a sample table.